FPT School of Business and Technology

FINAL PROJECT

APPENDIX I - API DOCUMENTATION

Course	:	DDM501.22 - AI in DevOps, DataOps, MLOps
Project topic	:	Predict Customer Satisfaction / Trust Toward AI in Online Shopping
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ML MODEL SERVING API — REFERENCE DOCUMENTATION

Base URL: `http://localhost:8000`

Interactive Swagger UI: `http://localhost:8000/docs`
ReDoc: `http://localhost:8000/redoc`
OpenAPI spec (YAML): `docs/openapi.yaml` — import into Postman / Insomnia / Swagger Editor

1 OVERVIEW

REST API that serves the **Credit Card Fraud Detection** classification model (RandomForest, binary: 0 = legitimate, 1 = fraud) registered in the MLflow model registry. All requests/responses are JSON.

Service	**URL**
API (this)	http://localhost:8000
MLflow UI	http://localhost:5001
Grafana	http://localhost:3000 (admin/admin)
Prometheus	http://localhost:9090
Evidently	http://localhost:8001
MinIO Console	http://localhost:9001 (minio/minio123)
Airflow	http://localhost:8080 (airflow/airflow)

2 ENDPOINTS

2.1 GET /

Service discovery. Returns available endpoint paths.

Response 200

```
{
  "message": "Credit Card Fraud Detection API",
  "version": "1.0.0",
  "endpoints": {
    "predict": "/predict",
    "explain": "/explain",
    "health": "/health",
    "metrics": "/metrics",
    "model_info": "/model/info"
  }
}
```

2.2 GET /health

Liveness + readiness check. Use for container health probes.

Response 200 — healthy

```
{
  "status": "healthy",
  "model_loaded": true,
  "model_name": "credit_fraud_model",
  "model_version": "4",
  "uptime_seconds": 3612.5
}
```

Response 200 — unhealthy (model failed to load)

```
{
  "status": "unhealthy",
  "model_loaded": false,
  "model_name": "credit_fraud_model",
  "model_version": "unknown",
```

```
    "uptime_seconds": 12.1
  }
```

2.3 POST /predict

Runs inference and returns whether the transaction is fraudulent.

2.3.1 Input features

The model expects **exactly 13 numeric features** in this order:

#	Name	Description	Typical range
0	`amount`	Normalized transaction amount	-3 – 5
1	`time_of_day`	Hour of day (standardized)	-2 – 2
2	`day_of_week`	Day of week (standardized)	-2 – 2
3	`merchant_risk_score`	Merchant fraud risk score	-2 – 4
4	`distance_from_home_km`	Distance from cardholder's home (standardized)	-2 – 5
5	`distance_from_last_txn_km`	Distance from last transaction (standardized)	-2 – 5
6	`ratio_to_median_amount`	Ratio of txn to median spend (standardized)	-3 – 5
7	`repeat_merchant`	Whether merchant was used before (standardized)	-2 – 2
8	`used_chip`	Chip was used for transaction (standardized)	-2 – 2

#	Name	Description	Typical range
9	`used_pin`	PIN was entered (standardized)	-2 – 2
10	`online_order`	Transaction is online (standardized)	-2 – 2
11	`foreign_transaction`	Transaction in foreign country (standardized)	-2 – 2
12	`txn_velocity_1h`	Number of transactions in last 1 hour (standardized)	-2 – 5

Output classes: `0` = legitimate, `1` = fraud

2.3.2 Request body

```
    {
      "features": [0.52, -0.31, 1.14, 0.78, -0.45, 0.93, -1.20, 0.38,
0.61, -0.77, 0.14, 0.89, -0.23],
      "feature_names": ["amount", "time_of_day", "day_of_week",
"merchant_risk_score",
                        "distance_from_home_km",
"distance_from_last_txn_km",
                        "ratio_to_median_amount", "repeat_merchant",
"used_chip",
                        "used_pin", "online_order",
"foreign_transaction", "txn_velocity_1h"]
    }
```

> `feature_names` is optional. Omitting it is valid; the API uses default fraud feature names.

2.3.3 Response 200

```
    {
      "prediction": 1.0,
```

```
      "model_name": "credit_fraud_model",
      "model_version": "4",
      "timestamp": "2026-03-23T10:15:30.123456",
      "latency_ms": 3.21
    }
```

2.3.4 cURL example

```
    curl -X POST http://localhost:8000/predict \
      -H "Content-Type: application/json" \
      -d '{
        "features": [0.52, -0.31, 1.14, 0.78, -0.45, 0.93, -1.20, 0.38,
0.61, -0.77, 0.14, 0.89, -0.23]
      }'
```

2.3.5 Error responses

Code	Cause
400	Non-numeric values or wrong feature count
503	Model not yet loaded (container starting up)
500	Unexpected server error

2.4 POST /explain

Returns **SHAP per-feature attribution values** for a single prediction.

Uses `shap.TreeExplainer` when available; falls back to global `feature_importances_` otherwise.

2.4.1 How to interpret results

- **Positive attribution** → feature pushes prediction *toward* fraud (class 1)
- **Negative attribution** → feature pushes prediction *away* from fraud (toward legitimate)
- `base_value` is the average model output (a.k.a. SHAP expected value)

2.4.2 Request body

Same schema as `/predict`.

```
    {
      "features": [0.52, -0.31, 1.14, 0.78, -0.45, 0.93, -1.20, 0.38,
0.61, -0.77, 0.14, 0.89, -0.23]
    }
```

2.4.3 Response 200

```
    {
      "method": "shap",
      "predicted_class": 1,
      "base_value": -1.2034,
      "attributions": {
        "amount":                        0.312,
        "time_of_day":                  -0.041,
        "day_of_week":                   0.008,
        "merchant_risk_score":           0.265,
        "distance_from_home_km":         0.195,
        "distance_from_last_txn_km":     0.143,
        "ratio_to_median_amount":        0.389,
        "repeat_merchant":              -0.015,
        "used_chip":                    -0.009,
        "used_pin":                     -0.022,
        "online_order":                  0.118,
        "foreign_transaction":           0.072,
        "txn_velocity_1h":               0.041
      },
      "top_features": [
        { "feature": "ratio_to_median_amount", "attribution": 0.389 },
        { "feature": "amount",                 "attribution": 0.312 },
        { "feature": "merchant_risk_score",    "attribution": 0.265 }
      ],
      "model_name": "credit_fraud_model",
```

```
      "model_version": "4"
    }
```

`method` value	Meaning
`shap`	Local SHAP values — sample-specific, recommended
`feature_importance`	Global RF importances — fallback when SHAP unavailable

2.4.4 cURL example

```
    curl -X POST http://localhost:8000/explain \
      -H "Content-Type: application/json" \
      -d '{
        "features": [0.52, -0.31, 1.14, 0.78, -0.45, 0.93, -1.20, 0.38,
0.61, -0.77, 0.14, 0.89, -0.23]
      }'
```

2.5 GET /model/info

Returns metadata about the currently loaded model.

Response 200

```
    {
      "model_name": "credit_fraud_model",
      "model_version": "4",
      "model_uri": "models:/credit_fraud_model/Production",
      "load_time_seconds": 1.23,
      "tracking_uri": "http://mlflow:5000"
    }
```

2.6 POST /model/reload

Hot-reloads the Production model from MLflow without restarting the container.

Response 200

```
{
  "status": "success",
  "message": "Model reloaded successfully",
  "model_version": "5"
}
```

2.7 GET /metrics

Returns Prometheus metrics in standard text exposition format.
Scraped every **15 s** by Prometheus.

Key metrics

Metric name	Type	Description
`api_requests_total`	Counter	Total HTTP requests — labels: `method`, `endpoint`, `status`
`api_request_latency_seconds`	Histogram	End-to-end request latency
`model_predictions_total`	Counter	Total predictions — labels: `model_name`, `model_version`
`model_prediction_latency_seconds`	Histogram	Inference-only latency
`model_prediction_value`	Histogram	Distribution of predicted class values (0/1)

Metric name	Type	Description
`model_prediction_errors_total`	Counter	Errors — label: `error_type`
`model_version_info`	Gauge	Currently loaded model version number
`model_load_time_seconds`	Gauge	Seconds taken to load the model
`model_feature_value`	Histogram	Per-feature input distribution (used by Evidently for drift detection)

Response — `Content-Type: text/plain; charset=utf-8`

```
    # HELP api_requests_total Total API requests
    # TYPE api_requests_total counter
    api_requests_total{endpoint="/predict",method="POST",status="200"}
142.0
    ...
```

3 PYTHON CLIENT EXAMPLE

```
    import requests

    BASE = "http://localhost:8000"

    # Fraud transaction features (standardized)
    features = [0.52, -0.31, 1.14, 0.78, -0.45, 0.93, -1.20, 0.38,
0.61, -0.77, 0.14, 0.89, -0.23]

    # Predict
    resp = requests.post(f"{BASE}/predict", json={"features":
features})
    print(resp.json())
    # → {"prediction": 1.0, "model_version": "4", ...}  (1 = fraud)

    # Explain
    resp = requests.post(f"{BASE}/explain", json={"features":
features})
    data = resp.json()
    print(f"Top feature: {data['top_features'][0]}")
    # → {"feature": "ratio_to_median_amount", "attribution": 0.389}
```

4 POSTMAN / INSOMNIA IMPORT

Import docs/openapi.yaml directly into Postman or Insomnia:

- **Postman**: File → Import → select `docs/openapi.yaml`
- **Insomnia**: Application → Import → From File → select `docs/openapi.yaml`
- **Swagger Editor**: paste contents at https://editor.swagger.io

5 ERROR SCHEMA

All errors follow the standard FastAPI error shape:

```
{
  "detail": "Human-readable error message"
}
```